Contact
LV Building Wire & Equipment Cables
Phone: 866-663-9267
USA.BuildingWire@nexans.com

ENERGEX® TFN and TFFN/AWM

PVC Insulated Cable 600 Volts - Copper Conductors - LEAD FREE

Description

Application

Nexans ENERGEX® TFN (solid) /TFFN (flexible stranded) copper conductors may be used as fixture wire, machine tool wire (MTW) or for appliance wiring material (AWM), as outlined in the NEC. These conductors are heat, moisture, and gasoline and oil resistant II.

TFN and TFFN are used for internal wiring of lighting fixtures and similar equipment, and for connection to branch circuit wiring inside lighting fixtures and similar equipment. TFN/TFFN is rated at a conductor temperature of 90ºC in dry locations. MTW is used for internal wiring of machines tools, rated 90ºC in dry locations and 60ºC when exposed to oils or coolants. AWM is used as factory installed internal wiring of appliances, rated 105ºC in dry locations.

Nexans TFN/TFFN conductors are Lead Free and RoHS compliant.

Standards

National UL 1063; UL 66; UL 758

Construction

Conductor:
TFN - solid annealed soft bare copper per ASTM B3.
TFFN/MTW/AWM - flexible stranded annealed soft bare copper per ASTM B174.

Insulation:
Color coded, heat, moisture and flame resistant polyvinyl chloride (PVC).

Jacket:
Strong, slick polyamide (Nylon) outer covering,

Colors:

Standard TFN and TFFN colors available in black, white, red, blue, green, yellow, pink, purple, orange, brown and gray. Additional colors are made per request subject to factory minimum order quantities.

Specifications

- Nexans ENERGEX ® TFN/TFFN meets or exceeds UL 66 and the requirements of the National Electrical Code - Fixture wire.
- TFFN meets or exceeds UL 1063 for Machine Tool Wire and UL 758 for AWM Style 1316. The insulation is heat, moisture, gasoline and oil resistant II.
- The TFN/TFFN shall be lead free and RoHS compliant. Voltage rating - 600 Volts.



Maximum operating temperature
90 °C

All drawings, designs, specifications, plans and particulars of weights, size and dimensions contained in the technical or commercial documentation of Nexans is indicative only and shall not be binding on Nexans or be treated as constituting a representation on the part of Nexans.

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Surface Print Legend (example)

NEXANS-F 16 AWG (1.31 mm2) (UL) TFFN or MTW or AWM 600V VW-1 GASOLINE AND OIL RESISTANT II LEAD FREE MADE IN CANADA (MFG. DATE)



Maximum operating temperature
90 °C

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Characteristics

Construction characteristics	
Conductor material	Copper
Electrical characteristics	
Maximum operating voltage	600 V
Usage characteristics	
Maximum operating temperature	90 °C

ENERGEX® TFN/TFFN Copper

Conductor Size AWG	PVC Insulation Thickness		Nylon Jacket Thickness		Approximate Cable Dimensions		Approximate Net Cable Weight		Ampacity* Amps	
	mils	mm	mils	mm	inches	mm	lbs/kft	kg/km	TFN/TFFN @ 90°C	MTW**
TFN										
18 (1)	15	0.38	4	0.10	0.081	2.05	7.1	10.6	6	
16 (1)	15	0.38	4	0.10	0.094	2.39	10.6	15.8	8	
TFFN / MTW										
18 (16)	15	0.38	4	0.10	0.087	2.21	7.4	11.0	6	7
16 (26)	15	0.38	4	0.10	0.100	2.54	11.0	16.4	8	10

* Allowable ampacities for TFFN and TFN are as shown in the National Electrical Code, Article 402.5.
**Allowable ampacities for MTW are as shown in NFPA 79.

Selling delivery information

Product Features

- Heat and moisture resistant
- Gasoline and oil resistant II
- Rated for use as appliance wiring
- Available from inventory in 11 colors
- LEAD FREE and RoHS compliant

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